

Computational Finance (WI4151) — Oral Exam: Worked Answers

Reconstructed from a 23 June 2026 exam report. Concept-level questions, English.

TU Delft — Prof. F. Fang | based on the course slides and assignments

Read this first — the exam format. The oral is *conceptual*: she asks about the *ideas* behind the methods, not about your specific code (you do *not* get to see your own code). A common opening: “*Which of the three numerical methods do you find easiest?*” — and you start there. So have a confident one-line summary of each method and be ready to go deep on the concepts. Everything below is the theory you need, in the order the questions were asked.

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1 The three numerical methods for option pricing

Q1. What are the three numerical methods for option pricing?

Monte Carlo (MC), the **COS method** (Fourier-cosine), and **Finite Differences (FD)**. (One can add the binomial tree and the Carr–Madan FFT, but these three are the headline methods.)

The unifying starting point is that *every* option price is a discounted risk-neutral expectation of the payoff,

$$V(x_0, t_0) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\text{payoff}(X_T)] = e^{-rT} \int_{\mathbb{R}} g(y) f(y | x_0) dy, \quad (1)$$

where X is the (log-)state, f is its risk-neutral transition density and g the payoff. The three methods are three different ways to evaluate this object:

- **MC** *samples* the expectation (1): simulate many paths of X , average the discounted payoff.
- **FD** solves the *pricing PDE* that (1) satisfies (Feynman–Kac), on a grid.
- **COS** evaluates the *integral* in (1) by expanding f in a Fourier-cosine series whose coefficients come from the *characteristic function*.

2 Main concepts of each method

Q2. For each method, explain the main concepts.

(She asks which you find easiest and you start with that. Below is the full content for each; lead with whichever you are most fluent in.)

2.1 Monte Carlo

Idea. Approximate $\mathbb{E}^{\mathbb{Q}}[g]$ by the sample mean over M independent simulated paths: $\hat{V} = e^{-rT} \frac{1}{M} \sum_{m=1}^M g(X_T^{(m)})$.

- **Why it is unbiased and how fast.** $\mathbb{E}[\hat{V}] = V$ (unbiased); the error is governed by the CLT, $\text{Var}(\hat{V}) = \sigma_g^2/M$, so the standard error decays like $O(M^{-1/2})$ — crucially *independent of the dimension* d . Report a confidence interval $\hat{V} \pm z_{\alpha/2} s/\sqrt{M}$.
- **Simulating X .** For GBM you can sample the terminal value *exactly* ($S_T = S_0 e^{(r-\frac{1}{2}\sigma^2)T + \sigma W_T}$). For a general SDE you time-step: *Euler–Maruyama* (strong order $\frac{1}{2}$, weak order 1) or *Milstein* (strong order 1, adds the $\frac{1}{2}bb'[(\Delta W)^2 - \Delta t]$ term).
- **Variance reduction** (same accuracy, fewer paths): antithetic variates ($\pm Z$), control variates (optimal coefficient $c^* = \text{Cov}/\text{Var}$, residual factor $1 - \rho^2$), importance sampling (change the sampling measure, reweight by the likelihood ratio), and quasi-Monte Carlo (low-discrepancy points, near $O(M^{-1})$).
- **Strengths / weaknesses.** Strength: high dimension and path-dependence are easy, the rate does not deteriorate with d . Weakness: slow $O(M^{-1/2})$ convergence; only a statistical error; early exercise needs extra machinery (Longstaff–Schwartz, Q3).

2.2 Finite differences

Idea. The price solves a PDE (Q10); discretise it on a grid and march *backward* in time from the payoff.

- Work in $x = \ln S$ so the Black–Scholes PDE has *constant coefficients*: $\partial_t V + (r - \frac{1}{2}\sigma^2)\partial_x V + \frac{1}{2}\sigma^2\partial_{xx} V - rV = 0$. Replace derivatives by difference quotients on a grid (x_j, t_i) .
- **Schemes** (see Q6): FTCS (explicit), BTCS (implicit), Crank–Nicolson; chosen for a stability/accuracy trade-off.
- **Boundary/terminal conditions.** Terminal condition = payoff; boundary conditions at the grid edges (e.g. Dirichlet). For *barriers*, a homogeneous (absorbing) boundary at the barrier. For *American/Bermudan*, impose the early-exercise constraint $V \geq \text{payoff}$ at every step (a linear complementarity problem, solved by projection / PSOR).

- **Strengths / weaknesses.** Strength: low-dimensional problems, American options (free boundary), and the Greeks read straight off the grid. Weakness: the curse of dimensionality — impractical beyond 2–3 state variables.

2.3 The COS method

Idea. Evaluate the integral (1) spectrally. Truncate the density to a finite interval $[a, b]$, expand it in a *Fourier-cosine series*, and note that the cosine coefficients are nothing but the *characteristic function* sampled on a grid. Pair them against the cosine coefficients of the payoff.

- Cost is a single $O(N)$ sum (no numerical integration, no FFT for the European case); convergence is *exponential* in N for smooth densities.
- Needs the characteristic function φ in closed form — true for Black–Scholes, Heston, the Lévy/jump models, and affine jump-diffusions in general.
- **Strengths / weaknesses.** Strength: extremely fast and accurate for European options, and it extends to Bermudan/American (Q4) and barriers. Weakness: needs the cf; effectively low-dimensional; you must choose the truncation range $[a, b]$ well (the one real pitfall, Q11).

The full algorithm is Q11; the “why it works” is Q7.

3 Bermudan options by Monte Carlo (Longstaff–Schwartz)

Q3. Can you price Bermudan options with Monte Carlo, and how?

Yes — by the Longstaff–Schwartz least-squares Monte Carlo (LSM) algorithm.

A Bermudan option can be exercised at a finite set of dates $t_1 < \dots < t_M$. At each date the holder compares the *intrinsic value* (exercise now) with the *continuation value* (keep the option), $V(t_m) = \max(\Lambda(S_{t_m}), c(S_{t_m}))$, where the continuation value is the discounted risk-neutral expectation of next period’s value,

$$c(S_{t_m}) = \mathbb{E}^{\mathbb{Q}}[e^{-r\Delta t} V(t_{m+1}) \mid S_{t_m}].$$

Why naive MC fails. Forward-simulated paths do not, by themselves, tell you the conditional expectation c at an interior node; estimating it by a fresh inner Monte Carlo at every node of every path is prohibitively expensive (nested simulation).

The LSM trick — regression. Replace the conditional expectation by a *least-squares regression* on a small set of basis functions of the current state (e.g. $1, S, S^2$, or weighted Laguerre polynomials). Working *backward* from t_M :

1. At t_M the value is the payoff.
2. At each earlier date t_m , take the *in-the-money* paths, regress the realised discounted future cash flows Y on the basis $\{\psi_j(S_{t_m})\}$, giving an estimate $\hat{c}(S_{t_m})$ of the continuation value.
3. Exercise on a path iff $\Lambda(S_{t_m}) > \hat{c}(S_{t_m})$; otherwise carry the cash flow forward.
4. At the end, discount each path’s single realised cash flow to t_0 and average.

Key points to state: regress only on *in-the-money* paths (the exercise decision is irrelevant elsewhere, and it conditions the regression where it matters); the regression *approximates* the conditional expectation; and the estimator has a small *low bias* (a sub-optimal exercise policy can only lose value) together with an in-sample “foresight” bias — so a clean estimate uses an independent out-of-sample set of paths for the valuation pass.

4 Bermudan options by the COS method

Q4. Can you price Bermudan options with the COS method, and how?

Yes — by the Fang–Oosterlee COS backward recursion.

Use the same dynamic-programming structure as LSM, but compute the continuation value by COS instead of by regression. In the log-state $x = \ln(S/K)$, for $m = M, M-1, \dots, 1$:

$$c(x, t_{m-1}) = e^{-r\Delta t} \int_{\mathbb{R}} v(y, t_m) f(y | x) dy, \quad v(x, t_{m-1}) = \max(\Lambda(x), c(x, t_{m-1})). \quad (2)$$

Each continuation integral is *exactly a European COS step*: with $F_k(x) = \text{Re}\{\varphi(\frac{k\pi}{b-a}; x) e^{-ik\pi a/(b-a)}\}$,

$$c(x, t_{m-1}) \approx e^{-r\Delta t} \sum_{k=0}^{N-1'} F_k(x) V_k(t_m),$$

where $V_k(t_m)$ are the cosine coefficients of the *next* step's value $v(\cdot, t_m)$. The work is to propagate those coefficients backward. Because $v = \max(\Lambda, c)$, at each date one finds the **early-exercise point** x_m^* where $c = \Lambda$ (by Newton's method) and *splits* the coefficient integral at x_m^* :

$$V_k(t_m) = \underbrace{G_k(\cdots)}_{\text{closed-form payoff part}} + \underbrace{C_k(\cdots)}_{\text{continuation part}}.$$

G_k is the analytic payoff coefficient (χ_k, ψ_k) ; C_k is built from the *previous* step's coefficients through a coefficient-coupling matrix that is the sum of a **Toeplitz** and a **Hankel** matrix — so each step is a discrete convolution computed by **FFT** in $O(N \log N)$. Total cost $O(M N \log_2 N)$. A Bermudan price with many exercise dates approximates the American (extrapolate in Δt).

Contrast worth saying out loud: the *European* COS method needs *no* FFT — it is one $O(N)$ sum. The *Bermudan/barrier* COS method *does* use the FFT, because at every exercise/monitoring date you must propagate the whole vector of cosine coefficients, and that propagation is a structured (Toeplitz + Hankel) matrix–vector product, i.e. a convolution.

5 Simulating two correlated sample paths

Q5. How do you simulate two correlated sample paths?

By the Cholesky factorisation of the correlation matrix.

You want two Brownian increments with $dW_1 dW_2 = \rho dt$. Draw two *independent* standard normals $Z_1, Z_2 \sim N(0, 1)$ and set

$$\boxed{W_1 = Z_1, \quad W_2 = \rho Z_1 + \sqrt{1 - \rho^2} Z_2.} \quad (3)$$

Then $\text{Var}(W_2) = \rho^2 + (1 - \rho^2) = 1$ and $\text{Cov}(W_1, W_2) = \rho$, as required. This is exactly the 2×2 Cholesky factor of the correlation matrix $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} = LL^\top$ with $L = \begin{pmatrix} 1 & 0 \\ \rho & \sqrt{1 - \rho^2} \end{pmatrix}$.

General case (n assets). Factor the correlation matrix $\Sigma = LL^\top$ (Cholesky, L lower triangular), draw an independent vector $Z \sim N(0, I_n)$, and set the correlated vector $W = LZ$; then $\text{Cov}(W) = LL^\top = \Sigma$. For increments over a step scale by $\sqrt{\Delta t}$.

Where it is used in this course: the two-asset basket Monte Carlo (assignment), and *Heston*, where the spot and variance Brownian motions are correlated (ρ , usually negative). One small subtlety to mention: the correlation is imposed on the driving Brownian *increments*; under a scheme like Euler this correlates the simulated state increments to first order.

6 Finite-difference schemes

Q6. Which finite-difference schemes are there?

Take the heat-equation form $u_t = u_{xx}$ (the Black–Scholes PDE reduces to it). With grid spacings $k = \Delta t$, $h = \Delta x$ and ratio $\nu = k/h^2$, the three standard schemes are:

Scheme	Type	Stability & accuracy
FTCS (forward time, central space)	explicit	conditionally stable, needs $\nu \leq \frac{1}{2}$; $O(k + h^2)$
BTCS (backward time, central space)	implicit	unconditionally stable; $O(k + h^2)$
Crank–Nicolson ($\frac{1}{2}$ FTCS + $\frac{1}{2}$ BTCS)	implicit	unconditionally stable; $O(k^2 + h^2)$

- **FTCS** updates each node explicitly from the previous time level — cheap, but a *von Neumann stability analysis* (insert a Fourier mode $u_j^i = \xi^i e^{i\beta jh}$) gives the amplification factor $\xi = 1 - 4\nu \sin^2(\beta h/2)$, and $|\xi| \leq 1$ forces the CFL condition $\nu = k/h^2 \leq \frac{1}{2}$. So refining the space grid forces a quadratically smaller time step.
- **BTCS** solves a (tridiagonal) linear system at each step; amplification $\xi = 1/(1 + 4\nu \sin^2(\beta h/2))$, so $|\xi| \leq 1$ *always* — unconditionally stable.
- **Crank–Nicolson** averages the two; it is second-order accurate in time and unconditionally stable — the usual default — though it can show mild oscillations near non-smooth payoffs.

(These are special cases of the θ -scheme: $\theta = 0$ FTCS, $\theta = 1$ BTCS, $\theta = \frac{1}{2}$ CN.) For *American* options the linear solve is replaced by a *linear complementarity problem* and solved with projected SOR (PSOR): project onto the payoff inside each sweep.

7 The main breakthrough of the COS method

Q7. What is the main breakthrough of the COS method?

That the Fourier-cosine coefficients of the density are obtained *directly and in closed form from the characteristic function* — no numerical integration and no Fourier inversion are needed.

In detail. On $[a, b]$ a function has the cosine expansion $f(x) = \sum_{k \geq 0} 'A_k \cos(k\pi \frac{x-a}{b-a})$, and the coefficient is the *inner product of f with the cosine basis function* (this is precisely the “ b_k via an inner product with cosine” your friend remembered):

$$A_k = \frac{2}{b-a} \int_a^b f(x) \cos\left(k\pi \frac{x-a}{b-a}\right) dx. \quad (4)$$

Now write the cosine as the real part of a complex exponential and *extend the integral to $\mathbb{R}$* (legitimate because f has negligible mass outside $[a, b]$):

$$A_k \approx \frac{2}{b-a} \operatorname{Re} \left\{ e^{-ik\pi \frac{a}{b-a}} \int_{\mathbb{R}} f(x) e^{i \frac{k\pi}{b-a} x} dx \right\} = \frac{2}{b-a} \operatorname{Re} \left\{ \varphi\left(\frac{k\pi}{b-a}\right) e^{-ik\pi \frac{a}{b-a}} \right\}, \quad (5)$$

because $\int_{\mathbb{R}} f(x) e^{iu x} dx = \varphi(u)$ is the characteristic function. So the inner product (4) equals the real part of the cf evaluated at the grid frequency $u_k = \frac{k\pi}{b-a}$.

Why this is the breakthrough. For almost all useful models the *density is unknown* but the *characteristic function is known in closed form*. Earlier Fourier methods (Carr–Madan) had to invert a Fourier integral numerically with an FFT and a damping parameter. COS replaces all of that with: evaluate the known cf at N points, take real parts, and sum against the (also closed-form) payoff coefficients. The result is an $O(N)$ algorithm with *exponential* convergence

for smooth densities. The structural reason it is so cheap is the orthogonality of the cosine basis (a Parseval/inner-product pairing): the integral of the product $f \cdot g$ becomes a simple sum of the products of their cosine coefficients.

8 Sampling from a CDF you cannot invert analytically

Q8. How do you sample a value when you have the (uniform) CDF but not an explicit inverse CDF?

By inverse-transform sampling with a *numerical* inversion of the CDF.

The inverse-transform principle: if $U \sim \text{Uniform}(0, 1)$ then $X = F^{-1}(U)$ has CDF F . When F^{-1} has no closed form, you do not need it analytically — for each draw U you *solve* the scalar equation

$$F(x) = U \tag{6}$$

for x with a root-finder: **bisection** (robust, F is monotone so the root is bracketed) or **Newton’s method** using the density $f = F'$ as the derivative, $x \leftarrow x - (F(x) - U)/f(x)$. Equivalently, evaluate F on a grid once and *interpolate* the inverse (build a lookup table $x \mapsto F(x)$ and invert by interpolation), which is efficient if you need many samples. (An alternative family is acceptance–rejection, when even F is awkward.)

Where this appears in the course — the Broadie–Kaya exact Heston simulation.

The hard step in sampling Heston exactly is drawing the *integrated variance* $\int_t^{t+\Delta} v_s \, ds$ given its endpoints. Its conditional distribution has no closed-form inverse, but its *characteristic function* is known; one recovers the conditional *CDF* from that cf (by Fourier/COS inversion), and then samples it by *numerically inverting the recovered CDF* (Newton on the COS-built CDF against a uniform draw). This is exactly “I have the CDF (from the cf) but not its analytic inverse, so I invert it numerically.”

9 Affine jump processes and their characteristic function

Q9. What is an affine jump process, what is the form of its characteristic function, and how do you get a and b ?

Definition. A process $X_t \in \mathbb{R}^d$ is an *affine jump-diffusion* (Duffie–Pan–Singleton) if its drift $\mu(X)$, its instantaneous covariance $\sigma\sigma^\top(X)$, the jump intensity $\lambda(X)$, and the discount rate $R(X)$ are all *affine* (linear + constant) functions of the state X :

$$\mu(x) = K_0 + K_1x, \quad (\sigma\sigma^\top)(x) = H_0 + H_1x, \quad \lambda(x) = l_0 + l_1x, \quad R(x) = R_0 + R_1x.$$

Black–Scholes (no jumps, log-price) and Heston (in $(\ln S, v)$) are affine; Merton/Kou jump-diffusions are affine.

Form of the characteristic function. For an affine process the (discounted) characteristic function / transform is *exponential-affine* in the state:

$$\varphi(u, \tau; x) = \mathbb{E}[e^{iuX_\tau} \mid X_t = x] = \exp(\alpha(\tau) + \beta(\tau) \cdot x), \quad \tau = T - t. \tag{7}$$

Here $\alpha(\tau)$ (the constant term, “ a ”) and $\beta(\tau)$ (the coefficient of the state, “ b ”) are functions of the time-to-maturity only.

How to get α and β — the Riccati ODEs. Substitute the ansatz (7) into the pricing/backward PDE that φ must satisfy (the Feynman–Kac / Kolmogorov backward equation for the affine generator, see Q10). Because every coefficient is affine in x , after dividing by φ you can *match* the terms that are *constant* in x and the terms that are *linear* in x separately. This collapses the PDE into a coupled system of ordinary differential equations in τ — *Riccati* equations (they are

quadratic in β because of the diffusion term $\frac{1}{2}\beta^\top H\beta$, and they carry a jump term $\lambda(\theta(\beta) - 1)$ where $\theta(c) = \mathbb{E}[e^{c \cdot J}]$ is the jump-size transform):

$$\beta'(\tau) = K_1^\top \beta + \frac{1}{2}\beta^\top H_1 \beta + l_1(\theta(\beta) - 1) - R_1, \quad \alpha'(\tau) = K_0^\top \beta + \frac{1}{2}\beta^\top H_0 \beta + l_0(\theta(\beta) - 1) - R_0,$$

with initial conditions $\beta(0) = iu$ and $\alpha(0) = 0$ (so that $\varphi(u, 0; x) = e^{iu x}$). You solve the β -Riccati first (often in closed form), then integrate to get α .

Concrete examples.

- **Black–Scholes** (no jumps): the ODEs give $\beta(\tau) = iu$ and $\alpha(\tau) = iu(r - \frac{1}{2}\sigma^2)\tau - \frac{1}{2}\sigma^2 u^2 \tau$, i.e. $\varphi(u, \tau) = \exp(iu(r - \frac{1}{2}\sigma^2)\tau - \frac{1}{2}\sigma^2 u^2 \tau)$.
- **Heston**: β for the variance state solves a scalar Riccati whose solution contains $D = \sqrt{(\kappa - i\rho\eta u)^2 + \eta^2(u^2 + i)}$ and $G = \frac{\kappa - i\rho\eta u - D}{\kappa - i\rho\eta u + D}$; α follows by integration. (Group the complex log / $\sqrt{}$ carefully — the “little Heston trap” — to stay on the right branch at long maturities.)

The payoff of all this: once you have α, β you have φ , and φ is the *only* model-specific input the COS method needs.

10 Deriving the pricing PDE

Q10. How do you arrive at the PDE for a pricing function?

Two equivalent routes; know both.

Route 1 — delta-hedging / replication. Hold the option and short Δ units of stock: $\Pi = V(S, t) - \Delta S$. Apply Itô to V (using $(dW)^2 = dt$):

$$dV = \left(\partial_t V + \mu S \partial_S V + \frac{1}{2}\sigma^2 S^2 \partial_{SS} V \right) dt + \sigma S \partial_S V dW.$$

Choose $\Delta = \partial_S V$ so the dW terms cancel: the portfolio is *instantaneously riskless*, hence by no-arbitrage it must earn the risk-free rate, $d\Pi = r\Pi dt$. Equating the two expressions for $d\Pi$ and cancelling gives the **Black–Scholes PDE**

$$\partial_t V + rS \partial_S V + \frac{1}{2}\sigma^2 S^2 \partial_{SS} V - rV = 0, \quad (8)$$

with terminal condition $V(S, T) = \text{payoff}$. Note the real-world drift μ *cancelled*: the price does not depend on it.

Route 2 — Feynman–Kac / martingale. Under the risk-neutral measure $\mathbb{Q}$ the discounted price $e^{-rt}V(S_t, t)$ is a *martingale*, so it has zero drift. Apply Itô to $e^{-rt}V$ and set the dt coefficient to zero — this gives the *same* PDE (8). More generally, the Feynman–Kac theorem says that for any diffusion X with infinitesimal generator $\mathcal{L}$, the conditional expectation

$$V(x, t) = \mathbb{E}[e^{-r(T-t)} g(X_T) \mid X_t = x] \quad \text{solves} \quad \partial_t V + \mathcal{L}V - rV = 0, \quad V(x, T) = g(x),$$

where for GBM $\mathcal{L} = (r - \frac{1}{2}\sigma^2)\partial_x + \frac{1}{2}\sigma^2\partial_{xx}$ in log-space (and for a jump model $\mathcal{L}$ gains a *non-local* integral term, giving a PIDE). This is the bridge between the expectation (1) and the PDE that FD solves — and it is the same backward equation whose exponential-affine solution gives the cf in Q9.

11 All the steps of the COS method

Q11. What are all the steps of the COS method (European option)?

1. **Write the price as a discounted expectation / integral.** $V = e^{-rT} \int_{\mathbb{R}} g(y) f(y \mid x_0) dy$, in the (log-)state y .

2. **Choose the truncation range $[a, b]$ from the cumulants** of the (log-)return:

$$[a, b] = \left[c_1 - L\sqrt{c_2 + \sqrt{c_4}}, \quad c_1 + L\sqrt{c_2 + \sqrt{c_4}} \right], \quad L \approx 8-12,$$

with c_1 the mean, c_2 the variance, c_4 the fourth cumulant. (For Black-Scholes $c_1 = \ln \frac{S_0}{K} + (r - \frac{1}{2}\sigma^2)T$, $c_2 = \sigma^2 T$, $c_4 = 0$.)

3. **Replace the truncated density by its Fourier-cosine series** on $[a, b]$: $f(y) \approx \sum_{k \geq 0} 'A_k \cos(k\pi \frac{y-a}{b-a})$.
4. **Get the coefficients A_k from the characteristic function** (the breakthrough, Q7): $A_k \approx \frac{2}{b-a} \operatorname{Re}\{\varphi(\frac{k\pi}{b-a})e^{-ik\pi a/(b-a)}\}$ — evaluate the known cf on the grid $u_k = \frac{k\pi}{b-a}$.
5. **Compute the payoff cosine coefficients** $V_k = \frac{2}{b-a} \int_a^b g(y) \cos(\cdot) dy$, which are *closed-form* for a call/put via the elementary integrals χ_k (of $e^y \cos$) and ψ_k (of $\cos$): e.g. call $V_k = \frac{2}{b-a} K [\chi_k(0, b) - \psi_k(0, b)]$.
6. **Insert the series, swap sum and integral** (justified by uniform convergence) and use orthogonality: $V \approx e^{-rT} \frac{b-a}{2} \sum_{k \geq 0} 'A_k V_k$.
7. **Substitute A_k and truncate at N terms** to obtain the COS pricing formula

$$V \approx e^{-rT} \sum_{k=0}^{N-1'} \operatorname{Re}\left\{\varphi\left(\frac{k\pi}{b-a}; x_0\right) e^{-ik\pi a/(b-a)}\right\} V_k. \quad (9)$$

8. **Check convergence.** Increase N ; for a smooth density the error falls exponentially and then *saturates at the range-truncation floor* set by $[a, b]$.

The one pitfall she loves (be ready for it). If $[a, b]$ is too narrow the price is wrong, and adding more terms N does *not* help — because that error is a *range* error, not a *series* error. This bites at *short maturity*: $c_2 = \sigma^2 T \rightarrow 0$, so the half-width $L\sqrt{c_2 + \sqrt{c_4}}$ collapses and $[a, b]$ becomes too small. Fix: use the cumulant rule (not a hand-picked window), keep the $\sqrt{c_4}$ term / a minimum width, and make $[a, b]$ cover the whole strike/spot grid.

One-sentence summary of the whole exam. Every price is $e^{-rT} \mathbb{E}^{\mathbb{Q}}[\text{payoff}]$; **MC** samples it, **FD** solves the PDE it satisfies, and **COS** evaluates the integral using the characteristic function — and the same characteristic function / affine structure, the same Cholesky correlation, and the same backward dynamic-programming recursion reappear across Bermudans, barriers, Heston and jumps.